MANUFACTURE AND TEST SEQUENCE

IN ORDER

1 — MACHINE

CP-101 body and CP-102 cover. Pocket, plenums, ports, mount holes.

2 — SKIVE FINS — one pass per fin

125 fins, 0.15 thk x 4.0 tall @ 0.40 pitch.

3 — DEBURR + CLEAN — no cutting fluid

Ultrasonic in alkaline, DI rinse, dry. Borescope every channel.

4 — BRAZE — fin tips NOT brazed

Vacuum or dry N2/H2. BCuP-5 preform on the perimeter land ONLY.

5 — PROOF PRESSURE — typically 15 bar

Hydrostatic 3 x working, 5 min, no permanent set.

6 — LEAK TEST — < 1e-8 Pa.m3/s

Helium mass spectrometer under vacuum.

7 — FLOW TEST — 6.1 kPa nom

Measure dP at 1.5 L/min. Record as the commissioning baseline.

8 — THERMAL TEST — first article + AQL

Die simulator at 700 W. Verify R <= 0.032 K/W.

9 — FLUSH + DRY — ship dry, capped

DI flush, N2 purge, cap both ports.

10 — INSTALL — air blankets fins

Mount per CP-200. Fill, vent, run 30 min, re-vent.

COOLANT SPECIFICATION

PROPERTY	SPECIFICATION	WHY
Fluid	**PG25, 25 % propylene glycol**	**freeze + biocide**
Inhibitor package	phosphate/azole, Cu-rated	protects copper
pH	8.0 - 9.5	copper passivity
Conductivity	< 50 uS/cm	galvanic / leak sensing
Particulate	filtered 50 um at the CDU	0.25 mm channels
Supply temperature	45 C nominal, 50 C max	warm-water loop
Flow per plate	1.5 L/min	design point
Change interval	per fluid supplier, test annually	inhibitor depletes

WETTED MATERIALS — GALVANIC COMPATIBILITY

MATERIAL	USE	VERDICT
Copper C10200, brass, bronze	plate, fittings	**ALLOWED**
Stainless 304 / 316	tubing, QD bodies	**ALLOWED**
EPDM	hose, external seals	**ALLOWED**
ALUMINIUM, any alloy	**anywhere wetted**	**PROHIBITED**
Zinc, galvanised steel	anywhere wetted	**PROHIBITED**
Nitrile, PTFE tape	seals	not recommended

Copper is strongly cathodic. One aluminium fitting in a copper loop corrodes

through in months and the debris then blocks the 0.25 mm channels.

COMMISSIONING

- Fill slowly from the lowest point. Vent at the highest point until clear.
- Run 30 minutes, re-vent, then record dP at 1.5 L/min as the baseline for this plate.
- Trend dP, not only temperature. Fouling shows in dP long before the die gets hot.
- Service trigger: dP 20 % above the commissioning baseline at the same flow.

ACCEPTANCE

TEST	LIMIT	APPLIED TO
Helium leak	**< 1e-8 Pa.m3/s**	**100 %**
Proof pressure	3 x working, no set	100 %
dP at 1.5 L/min	6.1 kPa +20 % / -30 %	100 %
Thermal R, die simulator	<= 0.032 K/W at 700 W	first article + AQL

INDIA-VC THERMAL HARDWARE — ISO 128 / ISO 2768-mK

TITLE: **PROCESS, TEST & COOLANT SPECIFICATION**

CP-100 COLD PLATE / MANDATORY SEQUENCE AND LOOP CHEMISTRY

SCALE: NONE — UNITS: mm — THIRD ANGLE PROJECTION

DRAWING No.: **CP-300** — REV: **A** — SHEET: **6 / 6**

DRAWN	CHECKED	MATERIAL	FINISH	DATE
AI-ASSISTED DRAFT	— PENDING —	SEE BOM	SEE NOTES	2026-09-10